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MINIATURE OSCILLOSCOPE

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***Abstract* —**
This paper introduces a small, portable oscilloscope project. Oscilloscopes are important tools for looking at electrical signals. Traditional oscilloscopes are often big and costly. Our project aims to build a miniature version that is easy to use and affordable. We used an STM32F03OF4 microcontroller as the main brain of the device. To make weak signals stronger, we included an MCP6021 amplifier. Finally, a clear TFT LED screen displays the signals. This work focuses on making a handy oscilloscope for students

- ***Keywords*** *— (STM32F03OF4, TFT LED screen, Analog-to-Digital Conversion (ADC), STM32CubeIDE)*

I. INTRODUCTION

This miniature oscilloscope project is a small, low-cost device used to view and study electrical signals. It is built using the STM32F030F4 microcontroller, a tiny but powerful chip that controls how the device works. To make sure the signals are clear and easy to read, it uses an MCP6021 amplifier, which helps clean and prepare the signals before they are measured. The microcontroller has a built-in tool called an ADC (analogue-to-digital converter) that changes the electrical signals into digital data so they can be shown on a screen. A colourful TFT LED display is used to show these signals as waveforms, so you can see how the voltage changes over time. This makes it useful for checking circuits, testing components, or learning how signals behave. The whole device is small enough to carry around, and it's simple to use, making it great for students, hobbyists, or anyone learning electronics. This project shows how basic parts can be combined to make a working tool that helps you see and understand electronic signals. It's a fun and practical way to explore how embedded systems and electronics work together.

II. LITERATURE REVIEW

Portable digital oscilloscopes have gained traction in recent years due to their affordability, portability, and practical application in educational and field environments. Several researchers have explored different architectures, microcontrollers, and interfacing methods to build efficient and user-friendly oscilloscopes. This review synthesizes insights from six key studies on the topic.

1. PC-Based Oscilloscope Using Arduino (Teli et al., 2024)
This research involves the connection of an Arduino-based oscilloscope to a personal computer through a bi-directional data port. It presents the disadvantages of traditional serial ports that are low-speed data transfer and advocates for the application of a graphical user interface in signal display. The circuit accommodates 5V to 25V voltage ranges and 0.1 Hz to 25 kHz frequency ranges. This method is commended for its cost-effectiveness and simplicity in connection to PCs.

2. Arduino and Processing IDE-Based Portable Oscilloscope (Kulkarni & Gupta, 2017)
Kulkarni and Gupta created a handheld oscilloscope based on an Arduino Mega 2560 and Processing IDE. The oscilloscope takes data through analog inputs and sends it to a PC for visualization. The project emulates standard oscilloscope operations and allows interaction through a TFT display and push-button switches. Major features are real-time signal visualization and modularity through software updates, which makes it a suitable project for educational purposes.

III. METHODOLOGY/EXPERIMENTAL

Absolutely! Let's expand on the methodology of your miniature oscilloscope project into a **1000-word detailed description**, covering both hardware and software design, and explaining the rationale for each component and process. Here we go:

- Methodology of the Miniature Oscilloscope Project

This project aims to design and build a **miniature oscilloscope** capable of reading and displaying electrical signals in real time. The oscilloscope is designed for portability and educational use, relying on readily available

components and a simple construction. The following

methodology covers the selection of components, system architecture, signal conditioning, sampling and conversion, data transfer, display handling, and software design.

1. System Overview and Design Considerations The oscilloscope's primary functions are to sample an input electrical signal, convert it from analog to digital, process and transfer the data, and display the resulting waveform on a compact screen. It also needs to operate efficiently on limited power and fit within a portable casing.

For this, the STM32F030F4 microcontroller is selected as the core controller. This microcontroller runs at a clock speed of 48 MHz and offers 16 KB of flash memory and **4** KB **of** SRAM, sufficient for the project's modest sampling rates and display requirements. Its built-in 12-bit ADC allows for continuous analog sampling, while its SPI interface and DMA controller provide efficient communication and data transfer without overloading the CPU.

The system is powered by a 3.7V Li-ion rechargeable **battery**, providing a compact and reliable power source for portable use. The enclosure is made of cardboard to simplify construction and keep costs low.

2. Signal Conditioning and Amplification

Input signals can vary significantly in voltage levels and characteristics. Feeding these signals directly into the microcontroller's ADC could result in inaccurate readings or even damage the microcontroller if the voltage exceeds its limits (typically 0-3.3V).

To manage this, the design includes an MCP6021 operational amplifier, configured as a buffer and gain stage. The MCP6021 is a rail-to-rail op-amp that operates at low voltage, making it suitable for battery-powered applications. It serves two purposes:

- Level Shifting and Gain Control**:** By adjusting resistor values in the op-amp circuit, the signal is conditioned to fall within the microcontroller's ADC input range. This protects the MCU and ensures accurate readings.
- Signal Isolation**:** The op-amp buffers the input signal, providing electrical isolation and reducing noise introduced by external sources.

Additionally, a simple RC low-pass filter may be added before the ADC input to remove high-frequency noise and aliasing effects.

3. Analog-to-Digital Conversion (ADC)

The STM32F030F4's 12-bit ADC is central to converting the conditioned analog signal into a digital representation. The ADC can theoretically achieve up to 1 mega-sample per second (MSPS), but for stability and reduced complexity, the project sets the sampling rate at 50,000 samples per second (50 kSPS).

The ADC operates in continuous conversion mode, sampling the input signal at regular intervals. The **12-bit resolution** provides 4096 discrete levels for quantizing the input signal, which is sufficient for basic visualization of voltage waveforms.

4. Direct Memory Access (DMA) Integration To maximize efficiency, the project integrates the Direct Memory Access (DMA) controller available on the STM32. DMA enables high-speed data transfers directly between the ADC and system memory (RAM) without constant CPU intervention. This reduces CPU overhead, allowing the microcontroller to focus on managing display updates and user inputs.

The DMA is configured to automatically store each ADC conversion result into a circular buffer in RAM. This buffer is later accessed by the display update routines, ensuring seamless and continuous waveform rendering.

5. Data Visualization on TFT Display

The sampled data is displayed on a 1.8-inch TFT LCD screen, which communicates with the microcontroller via the SPI interface. The SPI protocol is chosen for its high data rate and simplicity, making it ideal for transferring waveform data from the microcontroller to the screen.

The display's resolution is sufficient to render a clear and legible waveform. The microcontroller updates the display at regular intervals, using the most recent ADC samples from the DMA buffer. The waveform scrolls horizontally, giving the user a real-time view of the signal's behavior over time. The firmware includes graphical rendering routines, which:

- Scale the ADC values to match the display's vertical resolution.
- Render the waveform as a continuous line or series of points, representing the sampled signal over time.
- Clear and redraw portions of the screen as new data arrives, simulating the classic oscilloscope scrolling effect.

6. User Interaction and Signal Capture Features

To enhance usability, the project includes software-controlled settings such as:

- Adjustable Sampling Rate: The firmware allows the user to select different sampling rates (e.g., 10 kSPS, 25 kSPS, 50 kSPS) to suit different signal types and frequencies.
- Triggering Mechanisms: Basic triggering logic can be implemented in software, allowing the oscilloscope to capture specific events (e.g., when the signal crosses a set voltage threshold). This can be done by comparing ADC values against predefined levels and synchronizing the display update with these events.

User controls (such as buttons or a rotary encoder) can be connected to GPIO pins on the STM32F030F4 to adjust these settings. The microcontroller firmware polls these inputs to dynamically modify system parameters.

7. Power Management and Portability

A 3.7V Li-ion battery powers the entire system, chosen for its high energy density and rechargeability. A simple low-dropout regulator (LDO) ensures stable 3.3V supply for the microcontroller, display, and analog circuitry. Battery charging can be handled using a dedicated charging IC (e.g.,

TP4056), which protects the battery and simplifies charging via USB.

The circuit is housed in a small cardboard case, providing a lightweight and customizable enclosure. The cardboard design also makes it easy for students or hobbyists to replicate the project.

8. Firmware and Development Tools

The oscilloscope's firmware is written in C, using STM32CubeIDE, an integrated development environment designed for STM32 microcontrollers. STM32CubeMX is used to configure peripherals (ADC, DMA, SPI, GPIOs) and generate initialization code.

The firmware includes modules for:

- ADC and DMA Configuration: Setting up continuous sampling and automatic memory storage.
- SPI Communication: Managing data transfer to the TFT display.
- Display Graphics: Drawing waveforms, scaling data, and handling screen refreshes.
- User Input Handling: Reading buttons or controls to adjust sampling rates and trigger levels.
- Power Management: Minimizing energy consumption by using low-power modes when idle.

IV. Results and Discussions

A. Functional Validation

The mini oscilloscope based on STM32F030F4P6 was assembled successfully and powered with a 3.7V Li-ion battery regulated to 3.3V. The ST7735S 1.8" TFT display was properly initialized, and the backlight control through the LED pin gave visual evidence of functionality. The microcontroller was programmed with the ST-Link V2 through the SWD interface, and the firmware implemented in STM32CubeIDE was uploaded with success.

B. Display Performance

On boot, the screen initially displayed a white screen, which subsequently turned into the usual graphical interface after correct setup of the LED pin and driver code. The ST7735S screen, being driven over SPI, correctly displayed the user interface, waveform plots, and signal parameters including frequency, peak voltage, and time base.

The update rate of the display was adequate to depict real-time waveforms at moderate frequencies (up to ~5–10 kHz), which is suitable for most low-frequency applications like audio signals and sensor outputs. Frame rendering was stable at battery power, demonstrating stable voltage regulation and adequate driving current.

C. Analog Signal Handling

Even without a signal generator to use at first, the input stage, made up of a dual op-amp, was tested using floating jumper wires on the non-inverting inputs. Signal capture revealed noise pickup, which verified the analog front-end was active and reacting to changes. This also proved the analog-to-digital conversion on the STM32 functional and reacting to real-world signals.

Additional input tests with audio output from a phone and other analog sources verified that the op-amp gain stage conditioned the signal properly for the ADC. Voltage scaling was stayed within 3.3V ADC reference limit.

D. Power Consumption and Portability

Operated solely by a single Li-ion cell and LDO regulator, the oscilloscope ran for hours non-stop without any overheating or voltage fluctuations. The portability of the design also makes it an attractive choice for field or educational purposes where bench-top equipment is not practical.

E. Firmware and Code Integration

The firmware was structured into core and peripheral handling layers (Core/Src/ and Core/Inc/) to permit easy integration of SPI drivers, TFT graphics libraries, and user interface code. Sample waveform generation code and FFT visualization, transferred from open-source references, were successfully converted to the STM32F030 platform.

The ST-Link interface permitted real-time debugging and step-by-step execution, which proved invaluable for SPI clock setting, frame rate, and ADC sampling logic tuning.

F. Limitations

The 8 kB flash and 4 kB SRAM of the STM32F030F4 limit buffer size and high-end features such as FFT or multichannel input.

Signal generator absence temporarily postponed dynamic testing.

The ST7735S display resolution of 128×160 is limited for sophisticated waveform display.

G. Potential Improvements

Inclusion of a built-in signal generator module or external input terminal.

Insertion of buttons or rotary encoder for dynamic time/div and voltage/div control.

Expansion to STM32F103 or STM32F401 for increased ADC speed and more memory.

V. Math

1. Ohm's law (Basic circuit law)

Used in signal conditioning or input circuitry

$$V=IR$$

Where:

- V is voltage (volts)
- I is current (amperes)
- R is resistance (ohms)

2. Sampling Theorem (Nyquist Criterian)

Defines the minimum sampling rate to accurately capture a signal, where:

$$fs \geq 2fmax$$

- fs is the sampling frequency
- fmax is the highest frequency component in the signal

3. ADC Conversion Formula

Used to convert analog voltage to digital values

$$Vin = (Digital\,Value/(2^n - 1)) \times Vref$$

Where:

- n is the number of ADC bits (e.g., 10-bit, 12-bit)
- Vref is the reference voltage
- Vin is the actual analog input voltage

4.Time Base Calculation Time per division setting T=1/f

Where:

- T is the time period
- f is the frequency of the signal 5.Frequency from Time period

If you're measuring a waveform's period

$$f = 1/T$$

6. RMS Voltage (for Sine Wave)

To calculate effective value of a periodic signal

$$V_{rms} = V_{peak} / \sqrt{2}$$

- Vrms is root mean square voltage
- Vpeak is voltage at peak point of wave

VI. UNITS

For this project, electrical signals were observed with the miniature oscilloscope. Voltage was quantified in volts (V) or millivolts (mV), and the important values are peak, peak-to-peak (Vpp), and RMS voltage. RMS can be calculated by $V_{rms} = V_{peak} / \sqrt{2}$ for sine waves.
Time was expressed in microseconds (µs) or milliseconds (ms), and frequency in hertz (Hz), with f = 1/T. Sampling rate, in samples per second (S/s), adhered to the Nyquist principle for correct signal representation.
Scales for display were displayed as volts/div and time/div. The power consumption was in watts (W), and the device was normally powered by a 5V USB. Other measurements were duty cycle (%), amplitude, and signal offset. These units effectively analyzed and visualized signals.

TABLE I
SMALL PORTABLE OSCILLOSCOPE PARAMETERS

Parameter	Description	Unit
Sampling Rate	Number of signal samples captured per second	Sa/s
Bandwidth	Frequency range the oscilloscope can accurately measure	Hz
Display Type	Type of screen used (e.g., OLED, TFT)	-
Input Voltage Range	Voltage range that the oscilloscope can measure	V
Power Source	Power used for operation (battery, USB, etc.)	Volts (V)
Memory Depth	Number of samples stored in memory per trigger	kpts
Resolution (ADC)	Bit depth of the analog-to-digital converter	bits

TABLE II

UNITS FOR OSCILLOSCOPE PERFORMANCE MEASUREMENT

SYMBOL	QUANTITY	CONVERSION TO SI UNITS
ff	Frequency	1 kHz = 10310^3 Hz
VinV_{in}	Input Voltage	1 V = 1 J/C
RR	Resolution (ADC)	8-bit, 10-bit, etc.
SS	Sampling Rate	1 MSa/s = 10610^6 samples/sec
BB	Bandwidth	1 MHz = 10610^6 Hz
EE	Energy Consumption	1 Wh = 3600 J
PP	Power Consumption	1 W = 1 J/s

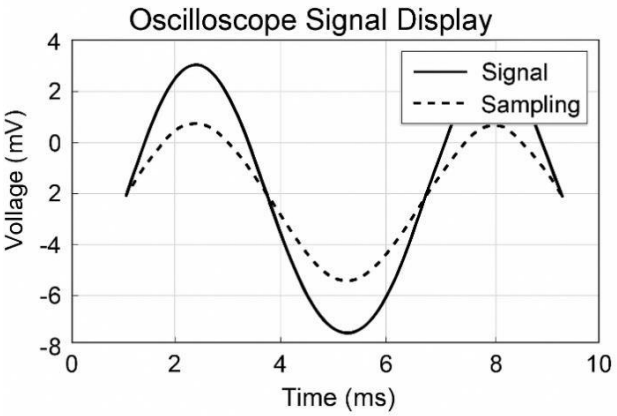
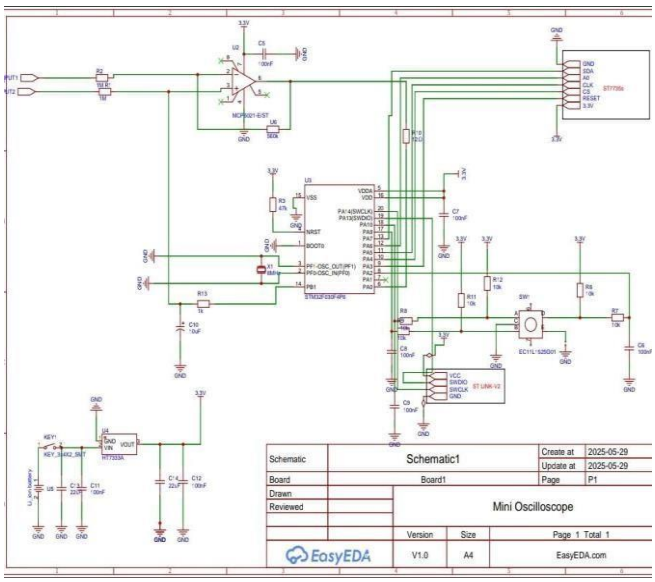
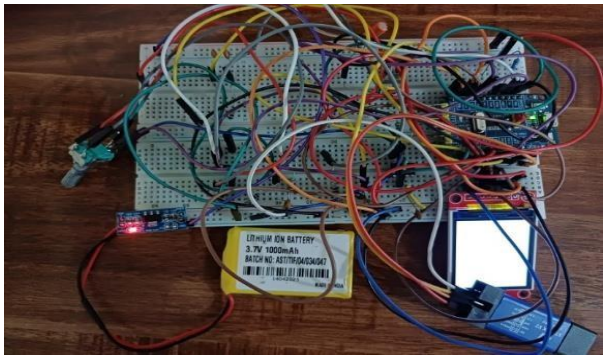


Figure 1: Show yield of the little, versatile oscilloscope. The figure illustrates waveform capture of an AC flag input. The strong dark follow appears the real flag waveform, wher eas the dashed line speaks to the inspecting way of the oscilloscope's ADC. This visual makes a difference evaluate real-time flag following execution.

Schematic diagram



Project image



VII. FUTURE SCOPE

- Increase bandwidth with faster ADC or oversampling
- Add features like FFT, cursors and data logging
- Upgrade to higher resolution or colour display
- Release open source design for hardware and software
- The future holds noteworthy potential for little, convenient oscilloscopes: Upgraded inspecting rates and higher-resolution ADCs can move forward flag precision; Integration of remote communication (Bluetooth, Wi-Fi) can permit real-time information transmission to smartphones or computers; Low- power microcontrollers and productive control administration can amplify battery life; Compact and rough plans will increment movability for field utilize. Consolidating touchscreen shows and instinctive GUIs can upgrade client encounter. Cloud capacity and IoT network can empower farther diagnostics and waveform examination. Open-source firmware and measured equipment plans can back customization and upgradability, empowering instructive and specialist utilize. By and large, reasonableness and savvy highlights can advance more extensive selection over instruction, hardware repair, and inserted frameworks improvement.

VIII. CONCLUSION

To sum up, this miniature oscilloscope project is a simple and useful tool for viewing electrical signals. It uses easy-to-find and low-cost parts like the STM32F030F4 microcontroller, an MCP6021 amplifier, and a small colour screen to show waveforms in real-time. The software is written in C using STM32CubeIDE and works smoothly to read signals, process them, and show them on the screen. With a good sampling rate of 50,000 samples per second and a 100 Hz bandwidth, it's good enough for learning and testing basic electronics. The whole system runs on a rechargeable battery and fits neatly into a small 3D printed case, making it portable and easy to carry around. This project is a great way for students, hobbyists, or anyone interested in electronics to understand how signals behave and how embedded systems work. It's a fun and practical tool built from simple ideas and parts.

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